

Deploying Deep Learning Burned-Area Detection Models on Sentinel-2 Imagery: A Case Study Comparing Swin-UNet and EfficientNet-B2 in Northeastern Portugal

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INTRODUCTION

Forest fires are among Portugal's most serious environmental and social issues each year. Every summer, large parts of the country experience fires, resulting in loss of life, destruction of homes and livelihoods, and damage to forests and woodlands. For instance, the June 2017 Pedrógão Grande fire killed 64 people, injured about 250, burned over 46,000 hectares in five days, and caused damage estimated at 200 million euros (Mayers, 2024). This event remains one of the most destructive recent fires. Portugal's vulnerability stems from a combination of climatic and land factors: hot, dry summers make the landscape prone to burning, and climatic conditions explain about 56% of the annual variation in burned areas (DaCamara, 2024). Years of rural abandonment have led to fuel buildup, and dense stands of fast-growing, highly flammable pine and eucalyptus trees enable fires to spread rapidly, with shrublands igniting under normal summer conditions and forests burning under extreme fire weather (Calheiros et al., 2022). Knowing the precise location and extent of a fire quickly influences many subsequent decisions. Remote sensing has become essential because it provides rapid, reliable assessments of burned areas, enabling prompt post-fire assessment (Llorens et al., 2021). Accurate burned area maps are crucial for guiding aerial firefighting, emergency logistics, and initiating legal and insurance processes.

The main objective of this study is to deploy and scale burned-area detection models to tile-wide inference, enabling the production of a continuous, georeferenced burned-area map for an entire Sentinel-2 scene. Specifically, it aims to:

1. Build an inference pipeline that converts the model's per-chip predictions into a single georeferenced burned-area map of the full Sentinel-2 T29TPG tile,
2. Tune the overlap aggregation scheme and a minimum-size filter to produce a seamless map free of small, scattered detections,
3. Compare the two fine-tuned models (MSR-BACD and EfficientNet-B2) against the ICNF fire records, including their runtime and memory costs, and recommend a practical deployment setting, and
4. Deliver a minimum viable web application that runs a chosen configuration and displays the resulting burned-area map on an interactive viewer.

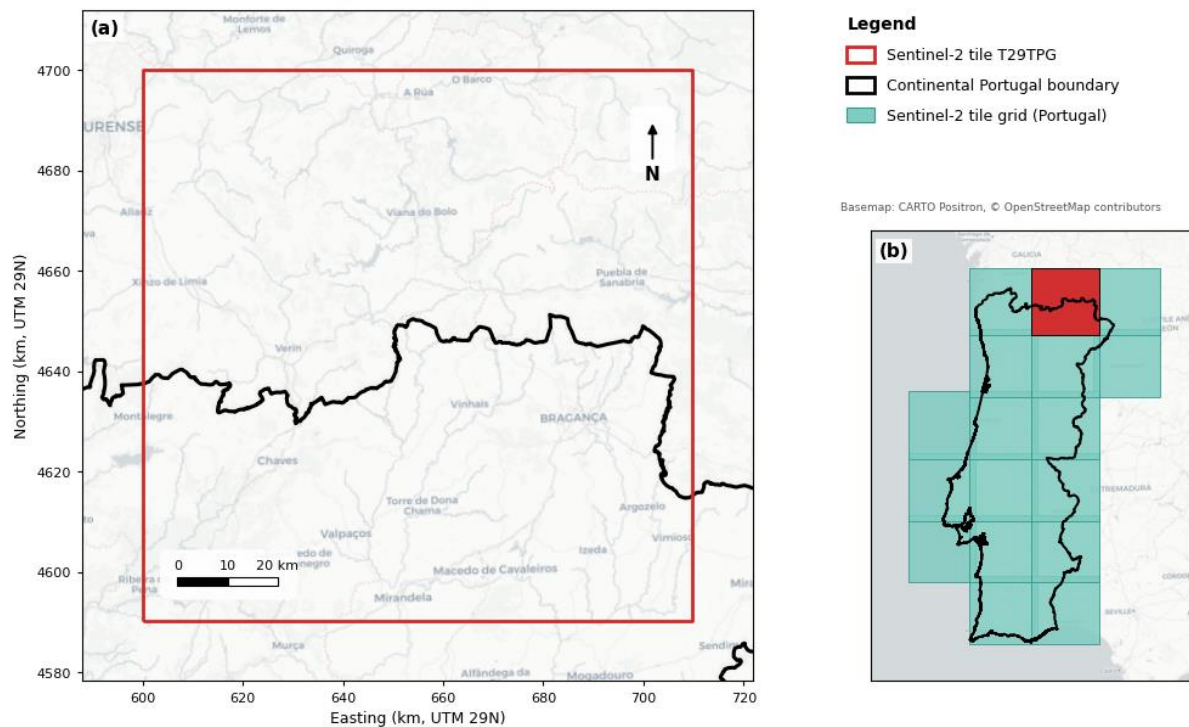
DATA

Study Area and Data Used

The study area focuses on Portugal's territory within the extent of Sentinel-2 Tile T29TPG in northeastern Portugal. The tile straddles the Portugal-Spain border and the study area corresponds to the Trás-os-Montes region in the districts of Bragança and Vila Real, a fire-prone landscape of shrubland, pine and oak forest stands, and agricultural areas, as shown in Figure 1.

Figure 1

Study area: Sentinel-2 tile T29TPG Extent in Northeastern Portugal



Note. The study area corresponds to Sentinel-2 tile T29TPG (red outline) in northeastern Portugal, shown in panel (a) with the continental Portugal boundary in black and in panel (b) relative to the Sentinel-2 tile grid covering the country; the analysis includes only the Portuguese territory within the tile. Basemap: CARTO Positron, © OpenStreetMap contributors.

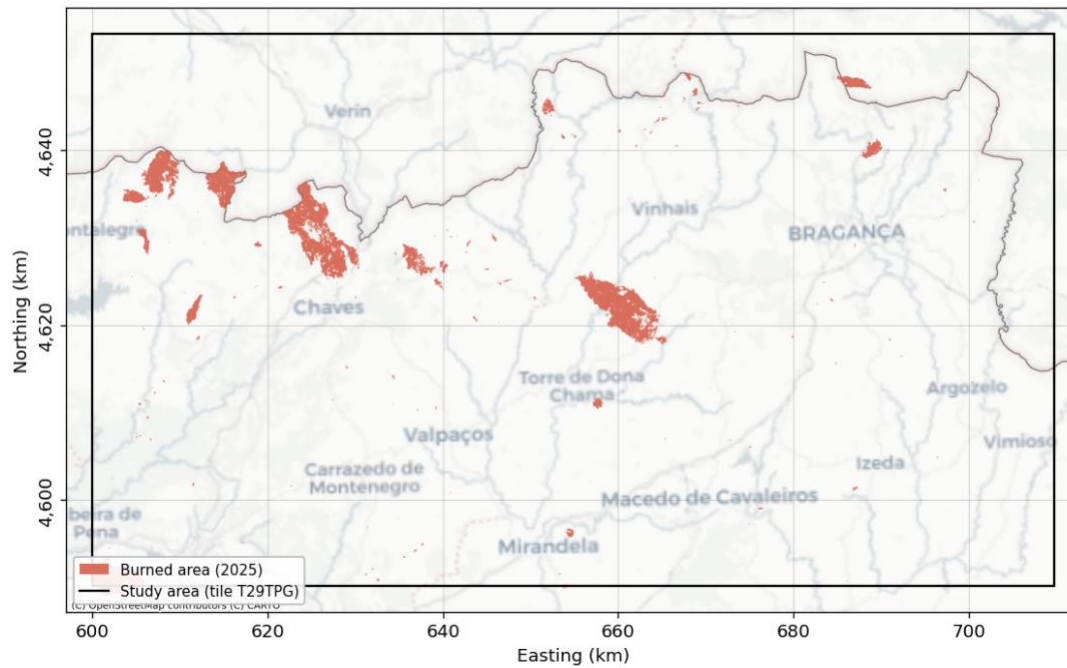
The project utilizes two existing datasets: Sentinel-2 imagery for northeastern Portugal in 2025 and ICNF's official burned-area records for that year. Sentinel-2 images serve as input to the model, while ICNF data provide fire perimeters for validation and event dates to select pre- and post-fire scenes. The input is a Sentinel-2 data cube for 2025, stored in a single HDF5 file, with the main array shaped (79, 10, 622,849,024), representing timestamp, bands, and pixels. This data cube includes 79 scenes acquired throughout 2025, roughly one every five days, each with ten

spectral bands covering visible, red-edge, and near-infrared regions at 10-meter resolution. Values are uint16 surface reflectance ranging from 0 to 10000, with NoData encoded as 65535. The array is chunked into shapes of (12, 10, 65,536), representing a 256 by 256 pixel chip with ten bands for 12 sequential scenes.

The reference data comprises the ICNF's official burned-area records for mainland Portugal in 2025 within the study area. Each fire in the shapefile includes the fire code, start and end dates and times, duration, district, municipality, parish, cause class, and burned area in hectares, categorized as forest, shrubland, or farmland. After masking the ICNF data to the study area, 182 fire events fall within the tile, of which 15 exceed 65 ha (Figure 2).

Figure 2

Spatial Distribution of 2025 Burned Areas within the Study Area



Note. Basemap: CARTO Positron, © OpenStreetMap contributors.

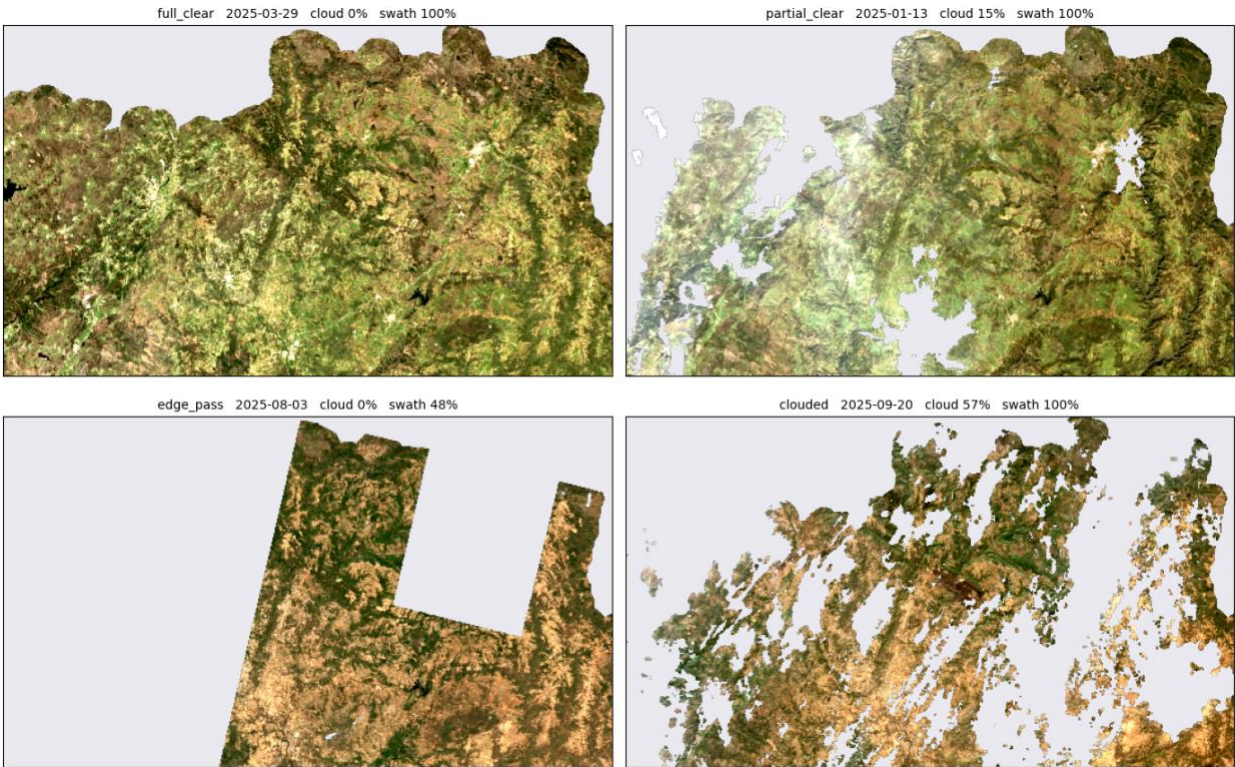
Data Cleaning

Cleaning is primarily done on the HDF5 file at the scene/timestamp level. Each acquisition is screened using two HDF5 metadata fields: swath coverage and cloud percentage. Using this field, the acquisitions are sorted into four categories as shown in Table 1. The example appearance of each tile in each category is shown in Figure 3. Only fully clear and partially clear scenes are eligible for inference; scenes that are under edge pass or clouded are dropped. All of the usable dates for the study are shown in Figure 4.

Table 1
Sentinel-2 Scene Quality Categories and Inference Eligibility

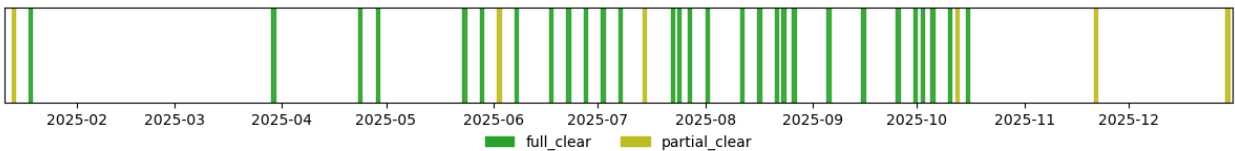
Category	Swath Coverage (%)	Cloud Cover (%)	Eligible for Inference
Full Clear	≥ 95	≤ 10	YES
Partial Clear	≥ 50	≤ 20	YES
Edge Pass	< 50	Any	NO
Clouded	≥ 50	> 20	NO

Figure 3
Representative Sentinel-2 Acquisitions for each Scene Category over Tile T29TPG, 2025



Note. Representative images are displayed as true-color mosaics (using bands B4, B3, B2 for R, G, B) at a 200 m resolution. Gray regions indicate areas outside the scene. A complete, clear scene fills the tile with unobstructed imagery; a partial, clear scene still covers the tile but includes clouds; an edge pass only clips a narrow strip; cloud-covered areas are mostly obscured. Panel labels specify the acquisition date, cloud cover, and swath coverage.

Figure 4
Temporal Availability of Usable Sentinel-2 Acquisitions, Tile T29TPG, 2025



DATA ORGANIZATION

Training and Validation

The two deep learning models in this study were developed independently and were not based on T29TPG. They were trained and tested using the VChips created. These VChips include 179 visual change-detection chips, each measuring 4 km by 4 km, with manual interpretation of before/after image pairs digitized in QGIS. They are categorized into five change classes (values 0 to 4) along with a separate NoData value (255). The chips cover five fire seasons: 24 from 2019, 81 from 2020, 26 from 2021, 37 from 2022, and 11 from 2023. An additional 42 VChips, taken outside the DGT potential-vegetation-loss mask, are labeled "no change" to help the models learn that normal changes in agriculture, montado, and urban areas do not indicate vegetation loss. Each chip includes its ten Sentinel-2 bands, ordered from B12 (longest wavelength) to B2 (shortest), plus a date band, which are reversed to the standard B2 to B12 order before training to match the prediction pipeline. For this study, the only class of concern is the burned areas (value 4), so every model output is reduced to a binary mask in which the fire class is treated as burned and all other classes as not burned.

Deployment Test Set

The test set includes all evaluable chips within tile T29TPG and is assessed against ICNF reference data to measure model accuracy (Chuvieco et al., 2019). There is no overlap between the 2025 evaluation and 2019–2023 VChip years. Ground-truth labels are derived from ICNF ardida_2025 polygons, rasterized onto the 10 m chip grid in the tile CRS (EPSG:32629), and eroded inward by one pixel to remove ambiguous burn-perimeter pixels, which could otherwise inflate error (Stehman & Foody, 2019). Chips in the pipeline that could not be observed are excluded from scoring. The evaluation uses a single shared before/after image pair across the entire tile, with a baseline date of 2025-07-07 and a subsequent scene of 2025-10-15, both clear and with 1% cloud cover.

METHODS

Models and Architecture

This study evaluates two fine-tuned change-detection models that share a common prediction interface and process the same ten-band Sentinel-2 chips before and after the event. The first model, Swin-YNet, is a fine-tuned MSR-BACD that employs a dual-temporal Swin Transformer encoder (Liu et al., 2021) and a U-Net-style decoder (Ronneberger et al., 2015). It produces three classes (0 Background, 1 Cuts, 2 Fires), with the Fires class representing burned areas. The second model is an EfficientNet-B2 U-Net (Tan & Le, 2019), featuring a pretrained EfficientNet-B2 encoder with about 10 million parameters. It takes the pre- and post- event images

stacked as inputs and outputs a binary label (0: Background, 1: Burned). Both models operate solely for inference, and each per-pixel output is converted into a binary burned mask using a model-specific burned-class index (Swin-YNet class 2, EfficientNet-B2 class 1).

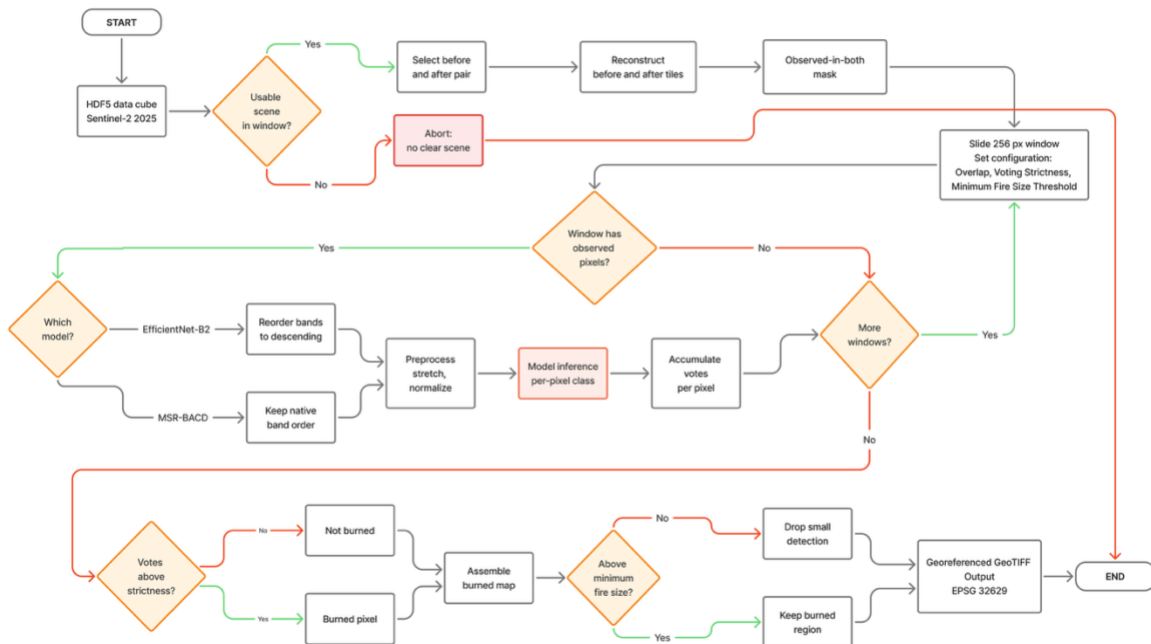
Inference Pipeline

Inference proceeds through four stages, transforming the raw HDF5 cube into a final map. First, scene screening labels each acquisition by swath and cloud cover, retaining only eligible scenes. Second, scene-pair selection selects a single global before/after pair per tile from specified date ranges, choosing the cleanest pre-season baseline and a late-season after-scene. Third, batched inference applies the chosen model to each data chip in batches of 8, reducing the outputs to the burned class. Fourth, mosaicking combines these per-chip results into tile-wide, georeferenced, single-band GeoTIFFs on the 10 m grid in EPSG:32629, producing three rasters: the multiclass prediction, the binary burned mask, and an observed mask indicating where both scenes have data, all compressed with LZW. Figure 5 shows the detailed flowchart of the inference pipeline.

Figure 5

Project Inference Pipeline Flowchart

Flowchart of Inference Pipeline



Validation

The models are validated on the burned class against the eroded ICNF reference labels. The supporting metrics are precision, the complement of commission error; recall, the complement of omission error; and their harmonic mean, F1, all calculated for the burned class only. The main single-number summary is the Matthews Correlation Coefficient (Matthews, 1975), selected because it considers all four cells of the confusion matrix and remains reliable under the heavy class imbalance typical of burned area mapping, where F1 and overall accuracy can be misleading (Chicco and Jurman, 2020). Overall accuracy is intentionally omitted because a simple classifier that predicts no burned pixels would still achieve above 95 percent (Stehman and Foody, 2019). Receiver operating characteristic and area under the curve are not used as primary metrics because predict returns a hard binary label rather than a probability (Fawcett, 2006). Instead, receiver operating characteristic and precision-recall curves are constructed from the vote fraction generated by overlapping windows, with the precision-recall curve serving as a more honest indicator in the presence of imbalance, and the full confusion matrix is provided alongside.

Post-processing Optimization

To optimize the inference model, three settings are tuned without changing the model weights. The first is the sliding window overlap (stride). Instead of using the cube's fixed, non-overlapping chips, a 256-pixel window moves across the tile with an overlap ranging from zero to seventy-five percent. Overlapping predictions are combined via majority vote, meaning a pixel is marked as burned if most covering windows agree. The second is voting strictness. The vote fraction (the percentage of windows that vote burned) acts as a threshold, defaulting to a simple majority (over fifty percent) and adjustable to balance false positives and omissions. The third is a minimum fire size filter that excludes connected burned areas smaller than a defined threshold, reducing speckle. By default, a twenty-five-pixel minimum patch size is used. The effects of each setting adjustment were evaluated.

Deployment

After identifying the optimal post-processing settings and running the inference model on the tile, a Streamlit app (`app.py`) was developed to transform those results into a minimal viable product (MVP). This MVP allows users to explore burned-area predictions interactively. Streamlit was selected for its suitability for quickly deploying machine learning models, as it encapsulates a Python inference pipeline within an easy-to-use web interface with minimal additional coding.

RESULTS

Comparison of Two Deep Learning Models

EfficientNet-B2 and Swin-YNet were tested on the same tile and paired images, then evaluated against the date-windowed ICNF ground truth. EfficientNet-B2 outperforms on all pixel metrics (see Table 2): precision of 0.56 versus 0.36, recall of 0.91 versus 0.85, F1 score of 0.69 against 0.51, and MCC of 0.705 compared to 0.542. Both models show higher recall with this date-windowed comparison than when tested against the full record, indicating many early misses were fires not visible as changes in either image. A McNemar test on disagreement pixels shows EfficientNet-B2 is correct 69.4% of the time, suggesting the difference is systematic. Error maps in Figure 6 reveal that Swin-YNet produces about twice as many false positives scattered across the tile, despite similar recall. Consequently, EfficientNet-B2 is chosen as the production model.

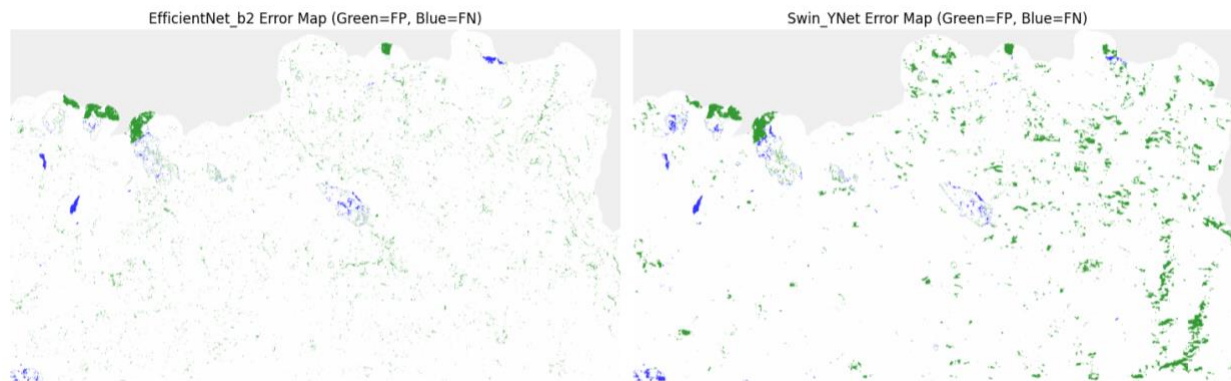
Table 2

Pixel-Level Metrics for EfficientNet-B2 and Swin-YNet against the Date-Windowed ICNF Ground Truth

Metric	EfficientNet-B2	Swin-UNet
True Positive	970,132	906,241
False Positive	774,989	1,580,949
False Negative	146,910	210,801
Precision	0.556	0.364
Recall	0.868	0.811
F1	0.678	0.503
IoU	0.513	0.336

Figure 6

Error Maps for EfficientNet-B2 (left) and Swin-YNet (right) against the Date-Windowed Ground Truth (Green Marks False Positives, Blue False Negatives)



Window Overlap and Voting Strictness

Four window overlaps were tested at 0, 25, 50, and 75 percent. MCC peaks at 50 percent overlap, because repeated voting across overlapping windows cancels scattered false alarms and lifts precision while recall stays high (Figure 7). Tightening the vote then helps further (Figure 8): at 50 percent overlap, raising the threshold from a simple majority to 75 percent lifts MCC from 0.70 to 0.76, and at 75 percent overlap the strictest setting tested, an 80 percent vote, reaches the highest value of the whole sweep at 0.80. The recall lost to stricter voting is small and spread evenly across fire sizes.

Figure 7

Precision, Recall, F1, and MCC for EfficientNet-B2 across the Four Window Overlaps Tested

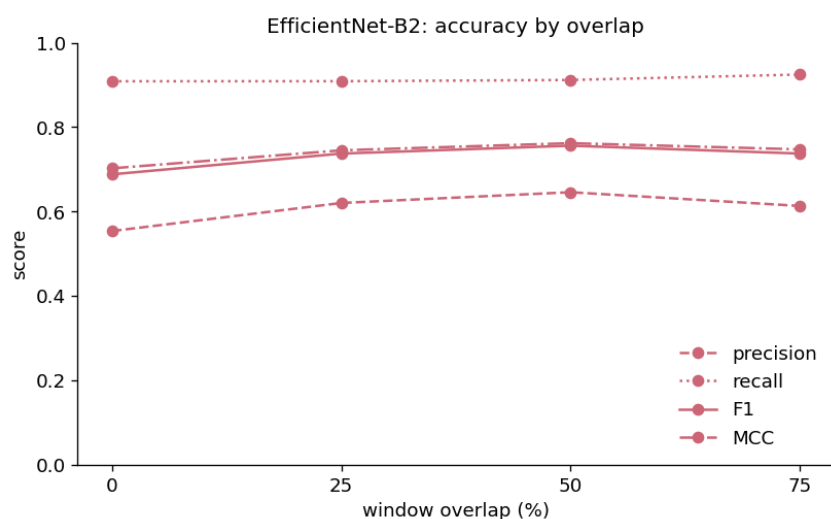
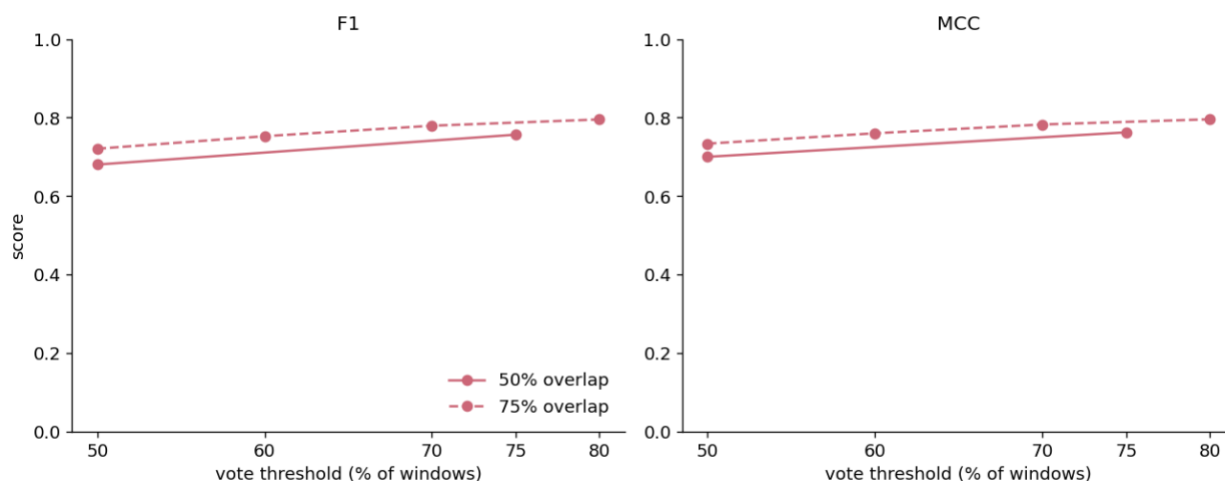


Figure 8

F1 and MCC against Vote Threshold, Compared at 50% and 75% Window Overlap



Minimum Fire Size Threshold

A minimum-size filter was then applied to remove small, scattered detections, sweeping the threshold from 25 to 975 pixels. Ranked by MCC alone, the filter keeps improving across the whole range, but this is misleading because the true fire sizes are dominated by many very small patches (Figure 9). Counted by patch, up to 92.7 percent of fires are classified as missed at the largest threshold, yet counted by area, only 2.34 percent of burned hectares are lost (Table 3). As Figure 10 shows, precision and MCC keep climbing while the true area lost stays under three percent; a per-class check, however, confirms that the filter also removes correct small-fire detections, not just noise. The threshold is therefore kept at 100 pixels, which equals one hectare and matches the ICNF sub-hectare boundary, so that the smallest fires the model can detect are protected.

Figure 9

True Fire-Size Distribution by Count (Bars) and Cumulative Area Burned per Bin (Line)

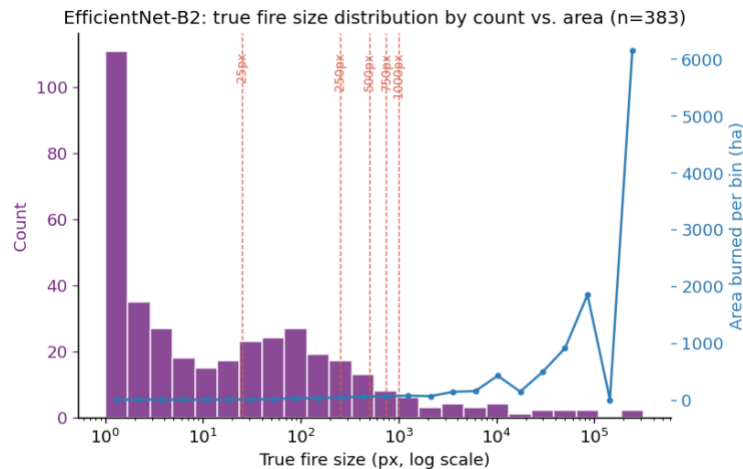


Figure 10

Precision, F1, and MCC against Cumulative True-Area Missed across the Minimum-Size Sweep

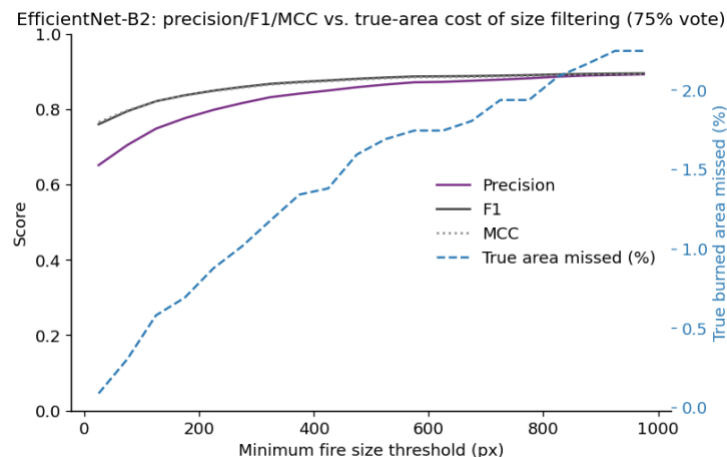
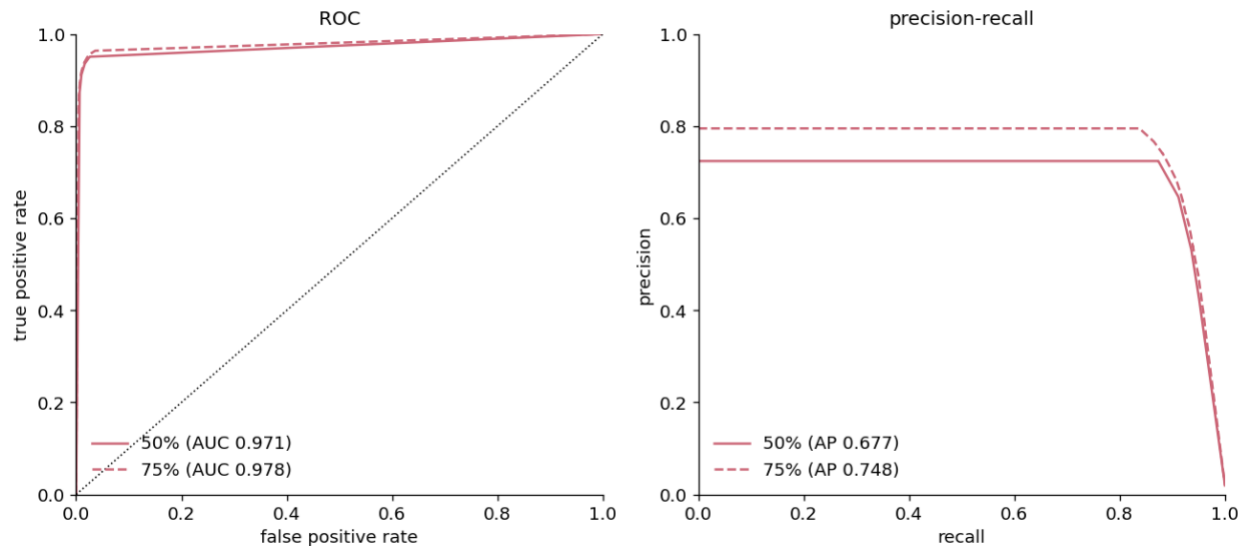


Table 3*Fires Missed by the Minimum-Size Filter, Counted by Patch and by Area*

Min size (px)	Precision	MCC	Fires missed by count (%)	Area missed (%)	Area missed (ha)
25	0.651	0.765	59.0	0.09	9.3
250	0.807	0.852	85.1	0.99	105.2
500	0.861	0.880	89.8	1.59	169.4
750	0.879	0.887	91.4	1.94	205.9
1000	0.892	0.893	92.7	2.34	248.8

ROC/PR Curves

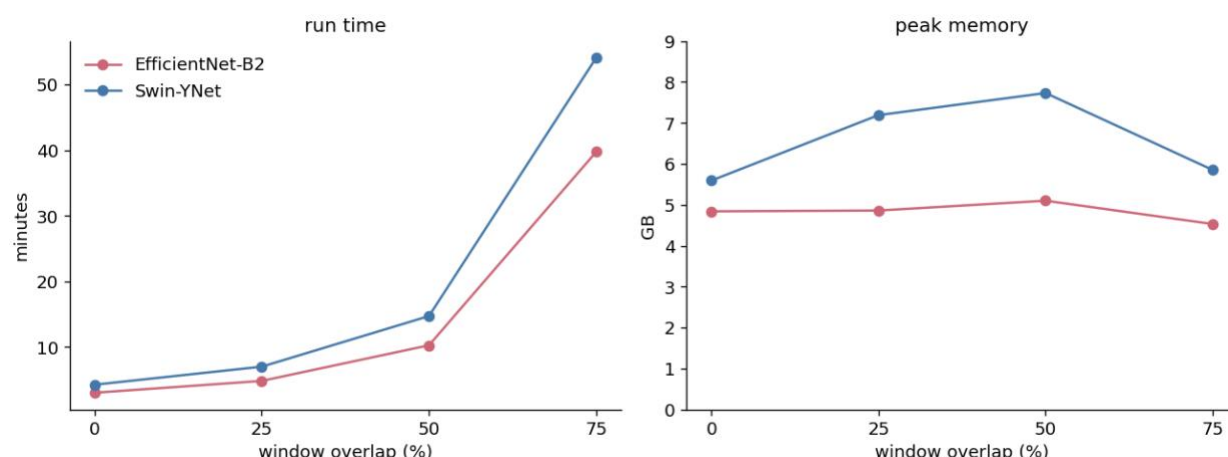
The ROC and precision-recall curves are derived from the raw vote fraction before any thresholds are set (see Figure 11). The ROC area is high at both overlaps, 0.97 and 0.98, but this figure can be misleading under severe class imbalance, as a large unburned background results in minimal changes to the false-positive rate. In contrast, average precision provides a more accurate reflection of performance, showing a broader and more sensitive range: 0.68 at 50 percent overlap and 0.75 at 75 percent, with the higher overlap producing a more distinct scoring separation.

Figure 11*ROC (Left) and Precision-Recall (Right) Curves Built from the Raw Vote Fraction, at 50% and 75% Window Overlap****Computational Resources***

Run time rises steeply with window overlap for both models, because the number of windows processed grows with overlap (Figure 12). EfficientNet-B2 climbs from roughly 3 minutes at no overlap to about 40 minutes at 75 percent overlap, and Swin-YNet from about 5 to about 55 minutes, so Swin-YNet is the slower model at every overlap. Peak memory follows a gentler pattern: EfficientNet-B2 stays close to 5 GB throughout, while Swin-YNet runs higher and peaks near 8 GB. This cost gap reinforces both the choice of EfficientNet-B2 and the use of a moderate 50 percent overlap rather than the far more expensive 75 percent setting.

Figure 12

Run Time and Peak Memory for EfficientNet-B2 and Swin-YNet across the Four Window Overlaps Tested



ANALYSIS

Which Model and Why

EfficientNet-B2 is the production model. It leads on every metric and on the date-windowed MCC (Table 2), the McNemar test shows the gap is systematic rather than noise, and it produces roughly half the false positives of Swin-YNet for comparable recall. The two are close on the largest, best-defined fires, but Swin-YNet's extra errors are scattered across the tile rather than concentrated at fire boundaries (Figure 6), consistent with its larger morphological closing radius, and they make its map noisier without improving recall. EfficientNet-B2 is also far cheaper to run. The one fire where both models fail together, Ribeira de Pena, is treated as an anomaly to inspect rather than as evidence against either model.

What is the Optimum Setting

The three post-processing levers do different kinds of work. Overlap buys precision almost for free, and the benefit has mostly arrived by 50 percent. Voting strictness buys further precision, and the finer resolution of 75 percent overlap only pays off once a strict vote is applied to it. Minimum size is the lever where following the headline metric blindly would mislead: ranking every configuration by MCC favours ever-larger filters (Table 4), because MCC is area-weighted and barely registers the loss of fires below about 20 hectares. Weighing detection against cost, the recommended operating point is 50 percent overlap, a vote of at least 75 percent, and a 100 pixel (one hectare) minimum size, which keeps almost all of the small-fire detection and runs at a fraction of the time of the highest-scoring but far more expensive 75 percent setting.

Table 4

Top Post-Processing Configurations Ranked by Matthews Correlation Coefficient

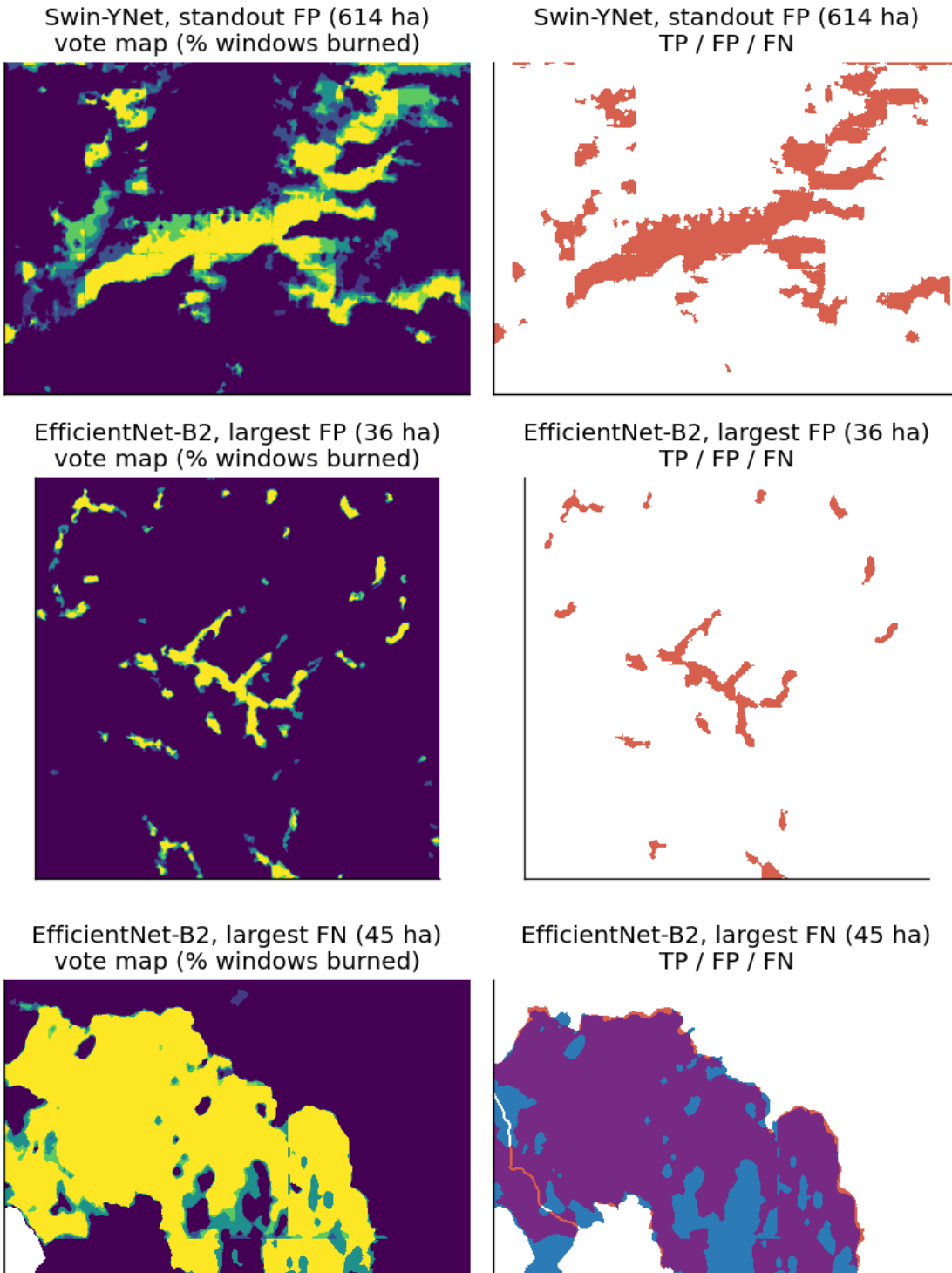
Rank	Setting	Burned area (ha)	Precision	Recall	F1	MCC
1	Overlap 50%, vote $\geq$ 75%, min size 975 px	10,701	0.892	0.898	0.895	0.893
2	Overlap 50%, vote $\geq$ 75%, min size 925 px	10,720	0.891	0.898	0.894	0.892
3	Overlap 50%, vote $\geq$ 75%, min size 875 px	10,738	0.889	0.898	0.893	0.891
4	Overlap 50%, vote $\geq$ 75%, min size 825 px	10,780	0.886	0.898	0.892	0.890
5	Overlap 50%, vote $\geq$ 75%, min size 775 px	10,845	0.882	0.899	0.890	0.888
6	Overlap 50%, vote $\geq$ 75%, min size 725 px	10,897	0.878	0.900	0.889	0.887
7	Overlap 50%, vote $\geq$ 75%, min size 675 px	10,946	0.875	0.901	0.888	0.886
8	Overlap 50%, vote $\geq$ 75%, min size 625 px	11,004	0.872	0.902	0.887	0.885

False Positive Analysis

A recurring error in both models is the confusion of river channels and their valleys for burned area. The standout false-positive clusters are shown in Figure 13 as vote maps alongside their true-positive, false-positive, and false-negative pixels. When the flagged area is checked against the Sentinel-2 imagery and an OpenStreetMap reference (Figure 14), the channel is visually unchanged between the two dates, while the surrounding farmland shows its own tonal changes consistent with ordinary land management rather than fire. Swin-YNet commits this error more strongly than EfficientNet-B2. Future training should include more examples of rivers and valleys, and in the meantime the app's focal-zone inspector lets a user pull up the before and after imagery to judge such cases directly.

Figure 13

Close-Up Comparison of Each Model's Standout False-Positive Cluster: Vote Map and TP/FP/FN Classification, for Swin-YNet's Largest False Positive (614 ha) and EfficientNet-B2's Largest (36 ha)



Sentinel-2 True-Colour Chip (After, 2025-10-15) and OpenStreetMap Reference for the River-Valley False-Positive Cluster

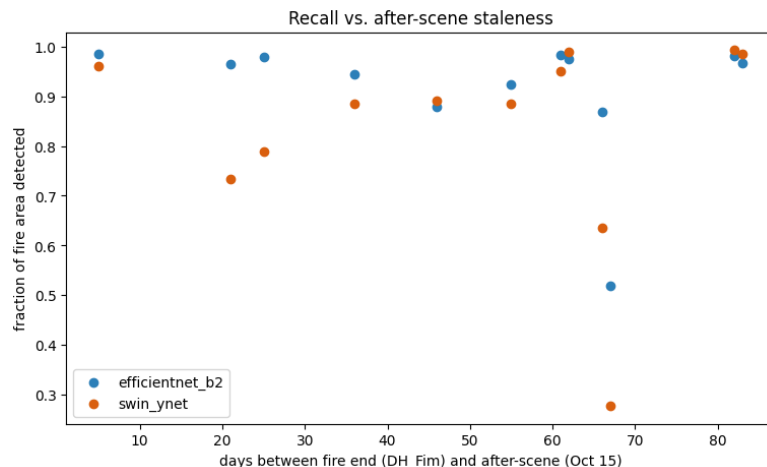


Limitations Discovered

Several choices limit how far these results generalize. Inference used a single before-and-after pair for the whole tile rather than one pair per fire, because repeating full-tile inference for every fire was judged too costly. The single after scene means 9 of the 12 evaluable fires are scored beyond the 30-day window the protocol specifies, although the recall of each fire shows no clear relationship with how stale its after scene is (Figure 15). Fire polygons were eroded by a fixed 10 meters with no sensitivity test. Scene eligibility was decided at the whole-scene level and so cannot catch localized cloud, and only fully clear scenes were used. Finally, the study covers a single tile and a single season, and the tile extends north into Spanish territory that ICNF does not map, so all scoring is restricted to continental Portugal.

Figure 15

Per-Fire Recall against Days between the Fire End Date and the After Scene (15 October 2025)



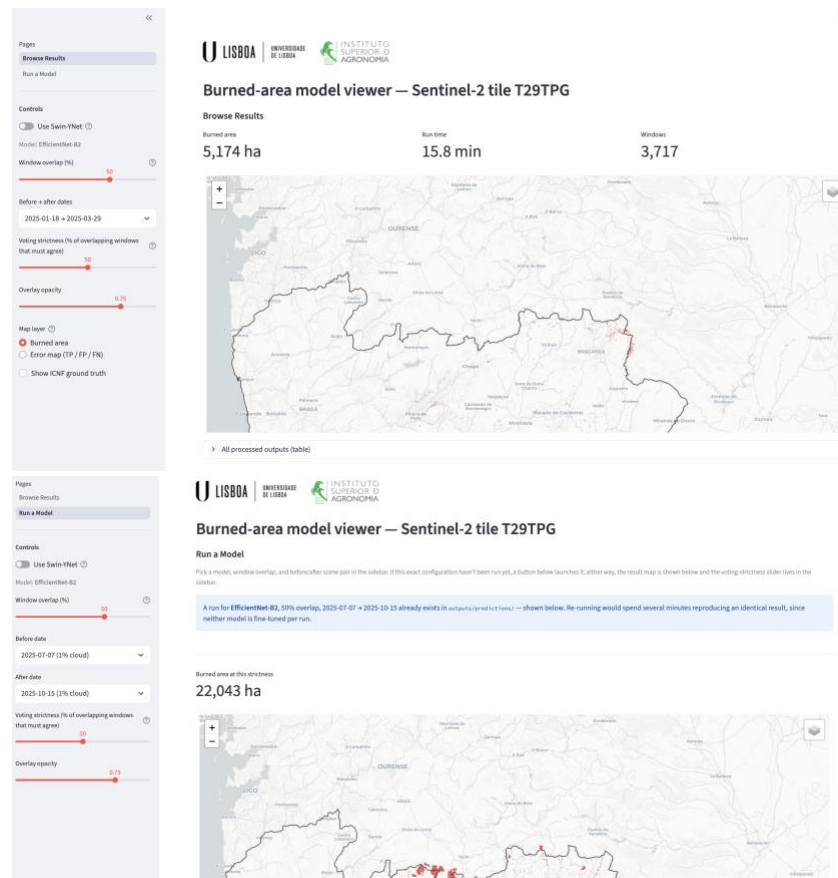
DEPLOYMENT

Once the best post-processing settings were identified and the models were run across the tile, the outputs were wrapped in a local Streamlit application so that a non-specialist can explore the burned-area predictions without touching the code. Streamlit was chosen because it is well suited to the fast deployment of machine learning models, making it a practical way to deliver a minimum viable product. The app is launched from the repository root, carries the Universidade de Lisboa and Instituto Superior de Agronomia branding, and is organized as two pages.

The Run a Model page lets the user pick a model, a window overlap, and a before-and-after scene pair. If that configuration already exists, it is shown at once; otherwise, a button launches the inference pipeline with a live progress bar. Either way, the result includes a voting-strictness slider that re-thresholds the map instantly without rerunning the model. The Browse Results page lists every run computed locally, switches between a burned-area overlay and an error map scored against ICNF, and provides a focal-zone inspector that, on a click, returns a small chip showing the Sentinel-2 before-and-after images, an OpenStreetMap reference, and that run's error map. Figure 16 shows the two pages of the app.

Figure 16

The Burned-Area Viewer: the Run a Model and Browse Results Pages



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